

Lecture 4 - Factorization into A=LU

→ Inverse of AB , A^{-1}

$$AA^{-1} = I = A^{-1}A$$

$$(AB)(AB)^{-1} = I = (AB)^{-1}(AB)$$

$$(AB)^{-1} = B^{-1}A^{-1}$$
$$(A(B B^{-1})A^{-1}) = I = (B^{-1}(A^{-1}A)B)$$

So, $\underline{\underline{(AB)^{-1} = B^{-1}A^{-1}}}$

→ A^T (square matrices)

$$(A A^{-1})^T = (I)^T$$

$$(A A^{-1})^T = I$$

$$\boxed{(A^{-1})^T} (A^T) = I$$

$$\underline{\underline{(A^T)^{-1} = (A^{-1})^T}}$$

- So basically transposing and inverting can be done in any order and they will lead to identical result.

→ Factorization into $A = \underline{L} U$

eg A

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} \xrightarrow{R^2 \rightarrow R^2 - 4 \cdot R^1}$$

$$\begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

E_{21} A U

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{matrix} L \\ \downarrow \\ E^{-1} \end{matrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

→ Inverse of E is reverse steps so

$$E^{-1} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} = L$$

$L \rightarrow$ lower triangular

$U \rightarrow$ upper triangular.

$\rightarrow$ Another thing we can do is to take out $\checkmark$ the factor and make the pivots of U as 1

$$\hookrightarrow \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}}_L \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}}_U$$

→ For 3×3 eg.

$$A = L U$$

$$E_{32} E_{31} E_{21} A = U$$

$$L = \underline{E_{21}^{-1} E_{31}^{-1} E_{32}^{-1}}$$

$$\rightarrow A = L U$$

if no row exchanges, the multipliers
go directly into L !!!

→ Basically it becomes super simple to construct L.

① start with identity

② Just keep putting the factors for creating pivot as zero.
(non-negative) - I mean opposite)

③ voilà

Just need to make sure there are no row exchanges and that they are done before.

→ How many operations on $n \times n$ matrix for elimination?

- Making every pivot 0.

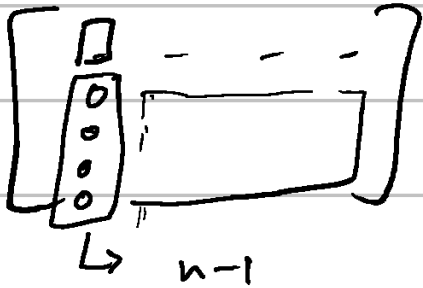
- If we count one row of as one op then at max
$$\frac{n(n-1)}{2} = \left(\frac{n^2 - n}{2}\right)$$

- Every row of would be n ops as $n \rightarrow$ multiply subtract op for each value

- So, for every pivot
$$\frac{n(n^2 - n)}{2}$$
 [don't know if correct]

→
$$\boxed{\frac{n^3 - n^2}{2}} \quad O(n^3)$$

→ Actually the correct way to look at it would be,



so $n(n-1)$ ops
 $n(n-2)$
 $\vdots$

$$\Rightarrow n(n-1 + n-2 + \dots + 1)$$

$$\Rightarrow \frac{n(n-1)(n)}{2} = \frac{n^3 - n^2}{2}$$

→ Why does gilbert & strat says the coefficient is $\frac{1}{3}$???

$$\rightarrow \text{Is } \frac{n^3 - n^2}{2} \approx \frac{1}{3} n^3$$

→ Row exchanges?

- What would happen if row exchanges were allowed.

- Using permutation matrices.

→ There are 6 permutation matrices of 3×3

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

→ It's a group!
Inverse is same except for cyclic. $P^{-1} = P^T$

→ # of $(n \times n)$ permutation matrices

$$n!$$